CprE 5360: Computer and Network Forensics

Potential Term Paper Topics

September 14, 2026

I. Requirements

Each 5360 student is required to finish a 6-page (single space and double column) term paper including defining a specific problem, surveying and comparing existing work, developing a (better) solution, and evaluating your results through experiments and/or analytical analysis.

You are welcome to propose your own term paper topic or can select a topic from the following list. The topic should be related to forensics at large.

Paper format should conform to the standard format for IEEE transactions. You can use either Latex or WORD. More information please refer: http://ieeeauthorcenter.ieee.org/create-your-ieee-article/useauthoring-tools-and-ieee-article-templates/ieee-article-templates/templates-for-computer-society-journals/. Please note that IEEE keeps changing the URL link. In case you find this is broken, you can go to www.ieee.org and search "Author Resource". Also, on Canvas, we have provided two zipped files for Word and Latex templates, if you cannot find them on IEEE Website.

II. Potential Topic List

The topics should be related to forensics in general, may include (but are not limited to):

- Detection of Deepfake, and other types of false evidence (e.g., forged, Planted, Tainted, Suppressed, or Parallel construction), as well as AI-enabled data (for misinformation and disinformation).
- Underground economic system analysis (spam email delivery, stock spam, botnets, click frauds, etc.)
- Solid State Drives (SSDs) (NAND flash memory) Imaging
- Mobile device forensics
- Device fingerprinting
- Triage Tool Analysis
- Network "neighborhood watch" systems
- Tracking illegal distributor in P2P systems
- Tracing anonymous VoIP calls and VoIP caller-ID services
- Tracking botnets: Membership and controllers/botmasters of Botnets
- Tracking attackers using Tor (http://tor.eff.org) and other stepping stones
- Tracking cyber criminals hijacking wireless networks (e.g., WIFI hotspots and mobile IP)
- Reconstructing Fragmented Videos (Some of the more common formats are MP4, AVI, and 3GP)
- Multimedia forensics and multicast fingerprinting
- Community structure analysis on social websites and Web 2.0 applications
- Bitcoin and Altcoins analysis and potential forensics issues
- IoT and vehicle forensics, for example, smart home devices, Apple AirTag, etc.
- Forensic tool validation: methodologies and principles, legal implications and issues
- Accountability and Fairness Verification on machine learning/unlearing

III. Deadlines

- September 20 (Sunday, 11:59pm), 1-page problem definition or topic description due.
- November 1 (Sunday, 11:59pm), 4-pages solutions and drafted evaluation plan due.

- December 6 (Sunday, 11:59pm), Final term paper due.